Opinion

Small island developing states

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In the international climate change negotiations, the Small Island Developing States (SIDS) have emerged as a credible group through the AOSIS (Alliance of Small Island States) and have called for a global temperature rise of 1.5°C above preindustrial levels. Whatever the outcomes of the negotiations, they are exacerbated by climate change and a rising sea level. Many share common characteristics of small size, high population density, limited land resources, vulnerability to natural hazards, threatened biodiversity, high dependence on tourism, and limited funds and human resources. The suggestions put forward for a research agenda for the SIDS include a comprehensive assessment of the SIDS as a group, a focused attention on oceans, increased development on renewable energy, inclusion of climate adaptation under natural disaster reduction, a strategy of 'save some islands rather than not to have any' for some SIDS, and large-scale modular mangrove planting for coastal protection and adaptation to sea-level rise. These suggestions could provide an expanded scope of adaptation for the SIDS. *WIREs Clim Change* 2011 2 1–6 DOI: 10.1002/wcc.84

INTRODUCTION

In the current climate change negotiations, one concern is to protect and safeguard the affected individuals and communities. In particular, one group of countries and territories, the Small Island Developing States (SIDS) contribute least to greenhouse gases but are most vulnerable to climate change. It should be noted that the SIDS are a highly diversified group as some islands that are not 'small' [e.g., Papua New Guinea (PNG)], not truly 'islands' (e.g., Guyana), not 'developing' (e.g., Bahrain), and not 'states' (e.g., Netherland Antilles).[1] Nevertheless, many share a distinctive character or uniqueness.[2] The small islands in the tropics have been popularized as 'islands in the sun' or 'pleasure islands' associated with their development for the global tourist market. The more familiar islands are in the Caribbean Sea, South Pacific, and Indian Ocean.

The progress of the SIDS in exerting their rights in climate negotiation and their vulnerabilities are briefly highlighted. Then the adaptation measures for the SIDS in their research agenda are examined. In addition to some measures currently in use or previously suggested, I will provide some new suggestions that would be helpful in future adaptation strategies and options.

In a dramatic address to the UN General Assembly on October 17, 1987, the President of the Maldives brought global awareness on the issue of climate change and the disappearance of small island states in face of a sea-level rise. Two years later, the country ran the first scientific conference on sea-level rise. In 1992, the world community adopted Agenda 21 and the small island states known subsequently as the Small Island Developing States (SIDS) were in the limelight of the first global conference on sustainable development and implementation of Agenda 21 held in April 1994 at Barbados. The Barbados Plan of Action recognized 14 areas of priority, including climate change; this plan was subsequently reviewed in January 2005 at Mauritius to produce the Mauritius Strategy Implementation. Within two decades, the small island states have moved from their images of pleasant tourist islands to sustainability challenges with climate change bringing heightened international attention.

Since 1990 the SIDS have emerged as a credible group known as the Alliance of Small Island States (AOSIS) in which the initial purpose was to unite them and increase their impact on climate change negotiations. At the 1994 Barbados meeting, the AOSIS emerged as a coordinating body for SIDS and all their

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DOI: 10.1002/wcc.84

interests. At the Bali climate change meeting of December 2007, Micronesia reaffirmed the AOSIS position that 'no island should be left behind'.[3] At Poznan in December 2008 when climate risk insurance was the buzz the AOSIS led the groups lobbying to ensure that insurance would become part of any deal on adaptation. On September 21, 2009 at the Climate Change Summit in New York the AOSIS adopted the Declaration of Climate Change that demands global average surface temperature increases to be limited to well below 1.5°C above preindustrial levels. At Copenhagen in December 2009, the AOSIS put up this temperature limit in their proposal and was supported by more than half of the members in the UN for the first time.

Progress in the road of climate change negotiations cannot hide the vulnerability of the SIDS. They account for less than 1% of greenhouse gas emissions, but are among the most vulnerable to climate change impacts and sea-level rise. The island chapters from the Intergovernmental Panel on Climate Change (IPCC) reports have clearly stated that climate change exacerbates the existing constraints and limited resources of the SIDS.[4,5] This is confirmed by other summaries of impacts and adaptation on the SIDS not listed in the latest IPCC report[1,6–8] and by various official websites (e.g., SIDNET, AOSIS, SPREP, CCCCC, CCRIF, US Fish and Wildlife Service Climate Change in the Pacific Region).

The vulnerability of the majority of SIDS to climate change and sea-level rise could be summarized in the following highlights.

- Limited physical space. Its small size, and sometimes consisting of a number of small islands, effectively limits adaptation option to climate change and sea-level rise.
- Small population but high density and increasing pressure on limited land resources. In some cases, the resources are heavily stressed from unsustainable human activities.
- Vulnerability to natural hazards, e.g., tropical cyclones (hurricanes), associated storm surge, droughts, tsunamis, and volcanic eruptions. In the Pacific islands natural disasters from 1950 to 2004 affected more than 3.4 million people with cyclones accounting for more than three-fourth of the disasters, outside PNG.[9]
- Relatively small watersheds and threatened water supplies. For many low-lying atoll states, their thin water lenses are highly sensitive to sea-level changes.
- Biodiversity is most threatened given their high degrees of endemism and levels of biodiversity (small size, isolated and fragility of island ecosystems).
- Narrow range of land resources. This forces undue specialization, e.g., tourism.
- Costly public administration and infrastructures (transport and communications), limited institutional capacities.
- Limited funds and human resource skills, limiting capacity to adapt to climate change.

In terms of vulnerability, the prospects of any sea-level rise based on the IPCC[10] estimates of 18–59 cm by 2100 would be dire to the low-lying nations of Tuvalu, Marshall Islands, Micronesia, and the Maldives and many atolls within other island nations. Various studies since the publication of the IPCC[10] report have pointed to a much higher sea-level rise by 2100. For example, at a conference to consider global consequences of climate change beyond 4°C held at the end of September 2009 at Oxford University, the scientists said that a 2-m sea-level rise is unstoppable over the centuries.[11] The sea level will continue to rise even with zero emissions now.

Sea-level rise depends to a large extent on global temperature increase, which contributes to thermal expansion of the oceans. Limiting global temperature increases has almost become the rallying point for the SIDS. The AOSIS has advocated the 1.5°C rise in its climate change negotiations and this has growing support from other nations. Limiting warming to below 1.5°C would require that concentrations of greenhouse gases in the atmosphere be limited to below 350 parts per million (ppm) of carbon dioxide equivalent.[12] It should be noted that while this and other dangerous limits of global warming are yet to have a strong scientific basis, there are reasons of concern[13,14] and indeed of more concern to the SIDS.

RESEARCH AGENDA

Apart from IPCC summaries, national reports, and regional studies and programs in the Caribbean and Pacific, there is no comprehensive review of present and projected climate change impacts, vulnerability, or adaptation for the SIDS region as a whole.[15] The Indian Ocean also lacks a regional program. There is a need to forge ahead with an assessment of all SIDS bringing together knowledge, experiences, data, data gaps, and future needs for the SIDS group. While it is a body for climate change negotiation, the AOSIS with the assistance from regional centers, international agencies, and non-governmental organizations (NGOs) could strengthen its role and heighten its profile with such an assessment. This would initiate a continued long-term research agenda for the SIDS.

Central to the SIDS is their relationship to the oceans and the seas. Many SIDS in the Pacific and Indian Oceans possess large EEZs (exclusive economic zones) extending up to 370 km. The oceans are, ironically, their biological and mineral resource and also their biggest liability. A chapter on oceans is in the next IPCC assessment report as we need to know more about the oceans in the global carbon cycle and the climate change and the resources which are vital to the SIDS. Increasing atmospheric carbon dioxide leads to ocean acidification that renders most regions of the ocean inhospitable to coral reefs, affecting tourism, food security, shoreline protection, and biodiversity. The impact of ocean acidification and how these impacts may propagate through marine ecosystems and affect fisheries remain largely unknown. They are to be reviewed once every 4 years by the Scientific Committee on Oceanic Research and Intergovernmental Oceanographic Commission which completed its second review in 2008. More than 60% of ocean acidification research papers have only been published since 2004.[16] While considerable attention has been paid to the coastal areas and we know much of their impacts of climate change, the SIDS need to pay more attention to the offshore and deep waters including sea-surface conditions and geophysical conditions. Continued sustainable exploitation and management of the living and non-living resources in large EEZs of the SIDS are the key to their development despite rising sea levels fueled by global climate change, marine pollution, and natural disasters such as tsunamis and hurricanes.

Although only a minor contributor to greenhouse gas emissions, the SIDS have taken action to reduce their overall emissions through an expansion of access to renewable energy and energy-efficient technologies—a demonstration for others to do so. Progress has been made in alternative energies, e.g., biofuels from sugarcane, energy from wastes using anaerobic digestion technique, solar energy, wind power, geothermal, and small hydropower. Other forms of renewable energy, e.g., ocean thermal energy conversion can also be competitive enough for the SIDS (OTEC website). Around 1997, the then president of AOSIS challenged the Climate Institute, a Washington DC-based NGO, to work with AOSIS, and the Global Sustainable Energy Islands Initiative consortium was formed in 2000 to accelerate the transition of up to 20 AOSIS member nations toward cleaner, more sustainable energy.[17] Perhaps more gauntlets need to be thrown at the NGOs and developed countries to get such programs and projects done!

Coastal hazards remain a constant threat to the SIDS and climate change adaptation (especially to sea-level rise) should include coastal hazards in regional and national programs. Empirical evidence of how islanders have responded and adapted in the past to sea-level changes[18] and to natural hazards such as earthquakes, volcanic eruptions, hurricanes, and tsunamis[19] provides valuable lessons to guide national planners on how to adapt to climate change. For the least developed countries in the SIDS, the National Adaptation Programme of Action is an easily understood process endorsed by the UNFCCC (United Nations Framework on Climate Change) to identify priorities for climate change adaptation.

Linking climate change to natural hazards faced by many of the SIDS is another aspect. Lessons learned from building resilience to storm risk can contribute to the creation of national level adaptive capacity to climate change, but climate change has to be prioritized before these lessons can be transferred. For SIDS, the governments play an important role in disaster risk reduction (DRR) and climate change adaptation. Governments could shape decision environment through appropriate laws and regulations; other approaches include assistance to people to develop causal links between environmental risks and impacts, a clear role for collective action, and creation of new institutional forms to manage risks. Embedding climate change in DRR would lead to a better perspective for improving research to serve policy and practice.[20] The lessons on post-tsunami coastal management[21] recommended adaptation to climate change to include reduction of coastal hazards within coastal zone management plans.

For some SIDS likely to suffer from a faster sea-level rise, it may be necessary to sacrifice a few islands now in order to save the others for the future. This is an extension of the 'Safe island' concept to a strategy of 'Save some islands'. The original safe island concept was used in the Maldives to relocate people from smaller islands to few larger islands with seawalls and amenities unavailable on smaller islands.[22] This concept was not new but failed due to the fear of destroying traditional village culture. Since the 2004 Indian Ocean tsunami, the concept has been expanded into a Safe Island Program in which islands have 'enhanced mitigation features' such as coastal protection, additional protection from vegetation, high grounds, and two-story buildings for vertical evacuation. I would suggest that the type of program could be further developed into a long-term strategy of 'Save some islands rather than not to have any' for atolls. Large-scale reclamation of selective islands to a higher elevation can be carried out by removing the material of other islands that would otherwise be drowned in the future. The technology for carrying such large-scale reclamation is not an issue as this is demonstrated in the islands reclaimed off Dubai. The main issue is funding and adaptation funds should be made available for such projects.

Similarly, the SIDS could expand and extend tested technologies that have been used or are familiar with, e.g., planting mangroves. Currently, mangrove planting is done by transplanting young seedlings in designated areas along the coast. This can be accelerated by mangrove planting on a large-scale modular system. Mangroves could be planted in modular units to reach required heights or maturity and the modular units be placed on the coasts that need protection. It is possible to add various sediments to increase the height of the mangrove area as sea level rises. Field observations by the author in various parts of Southeast Asia show that mangroves can grow on substrates ranging from mud, sand, gravels, boulders, coral flats, limestone platforms, and even on riprap surfaces of coastal protection measures. Thus, the assembly of modular units and the addition of sediments and appropriate nutrients are the key elements to this strategy of coastal protection.[23] As mangroves take time to grow and for islands where sea-level rise is more gradual, this technology would be suitable. This methodology is encouraged by recent evidence that the coral islands in Pacific Ocean are able to keep up with a rising sea level provided sediments are available.[24]

Although factors other than climate change played a role in inter-island and atoll migration in the Pacific, the media and academic literature have speculated that climate change may force people to migrate from their island homes.[25] In November 2008, the President-elect of the Maldives announced a plan to divert a portion of the country's billion-dollar annual tourist revenue into buying a new homeland as an insurance policy against climate change that threatens to turn the 300,000 islanders into environmental refugees. Whether islanders are moving into new homes or forced as climate refugees, this has serious international implications as host countries have to be prepared for the refugees. But islanders like most people do not wish to lose their cultural identity and community and remain for reasons of lifestyle, culture, and identity.

CONCLUSION

Although less than 20 years ago the SIDS were accorded the rights to sustainable development, their plight epitomizes the central challenge implicit in the construct of equity as defined by the Brundtland Commission. An interesting principle was in the AOSIS Dialogue Working Paper No. 14, August 2007 at Vienna[26]: the 'avoidance of climate change on SIDS' as a benchmark for post-2012 negotiations. A year later, the above principle was adopted by the UN Human Rights Council as a rights approach to climate change—a vital step in increasing global urgency to the climate change matter. It is something new and not in the UNFCCC and the SIDS have a good case.

Yet the majority of the SIDS face hard facts given the prospects of a rapidly sea-level rise. Even zero emissions would not improve the plight of the SIDS as there is a lag time before sea level stops rising. In their study, Nurse and Moore[27] concluded that adaptation is not just an option but an imperative for the SIDS. I hope that the various suggestions provide a wider scope and stronger base for the imperative adaptation of the SIDS.

ACKNOWLEDGEMENTS

The author is grateful to the two anonymous referees for their comments.

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